

Carbon Dioxide Projections

Time Series Analysis of CO₂ in Niwot Ridge, Colorado from 2012 through 2021

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9 December 2022

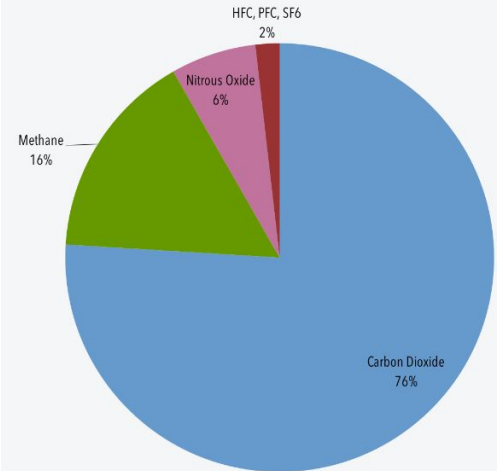
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Rationale

- Greenhouse gas emissions occur daily and have many detrimental effects
 - Global warming
 - Rising temperatures, and thus rising sea levels
 - Destruction or alteration of ecosystems
- Carbon dioxide is by far the most often emitted greenhouse gas, making it arguably the most important one to understand
- Climate change affects everyone and everything
- Understanding the past carbon dioxide patterns in order to predict the future
- Predictions can be used to hypothesize quality of life for humans, animals, and nature alike

Global Manmade Greenhouse Gas Emissions
by Gas, 2015



Dataset

- Collected by the National Oceanic and Atmospheric Administration
 - Using non-dispersive infrared (NDIR) analyzer
 - Calculated dry air mole fractions of carbon dioxide
- Niwot Ridge, Colorado from 1967 through 2021
- Initial Observations
 - Missing values between 1974 and 1975
 - Complete from 1976 through 2021
 - No major outliers, generally consistent patterns
- Data from 2012 through 2021 exhibits similar patterns to other subsets of data
 - Represents more recent and thus more useful data
 - Maintains a relatively large sample size of 120 time points
 - Stationarity likely attainable
 - Reasonable to assume patterns will continue
- Splitting the data set
 - 90% or nine years, 2012 through 2020, used for training data to develop each method
 - 10% or one year, 2021, used as testing data for each forecasting method



Exploratory Data Analysis

- Clear linear trend with a constant and positive slope
- Clear seasonality with peaks around March and April and valleys around August and September each year
- Does not appear to be a significant amount of random noise or fluctuations away from the general pattern, i.e. the pattern is consistent
- Reasonable to assume that patterns will continue into the future

Fig. 1 - Carbon Dioxide in Colorado (Niwot Ridge)

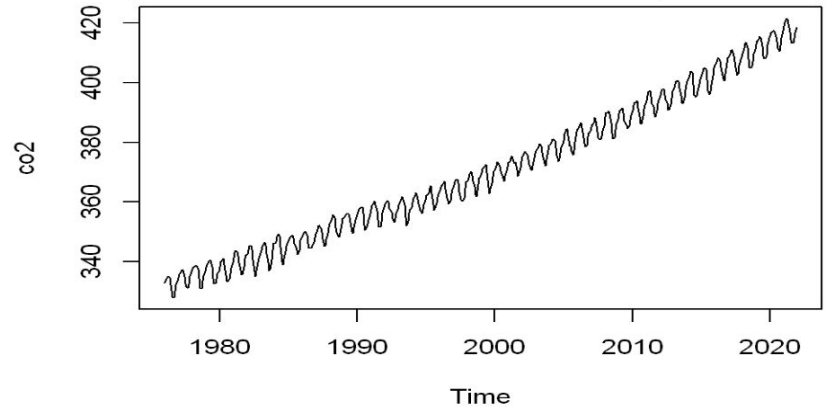
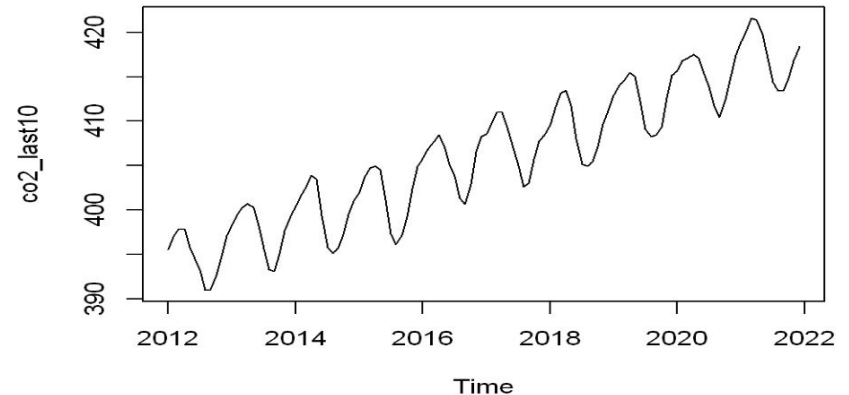


Fig. 2 - Carbon Dioxide in Colorado (Niwot Ridge)



Time Series Regression

A time series regression model is written with the predictive variable (y) on the left side and as a function of the explanatory variables and an error term to the right side to ensure random fluctuations are accounted for.

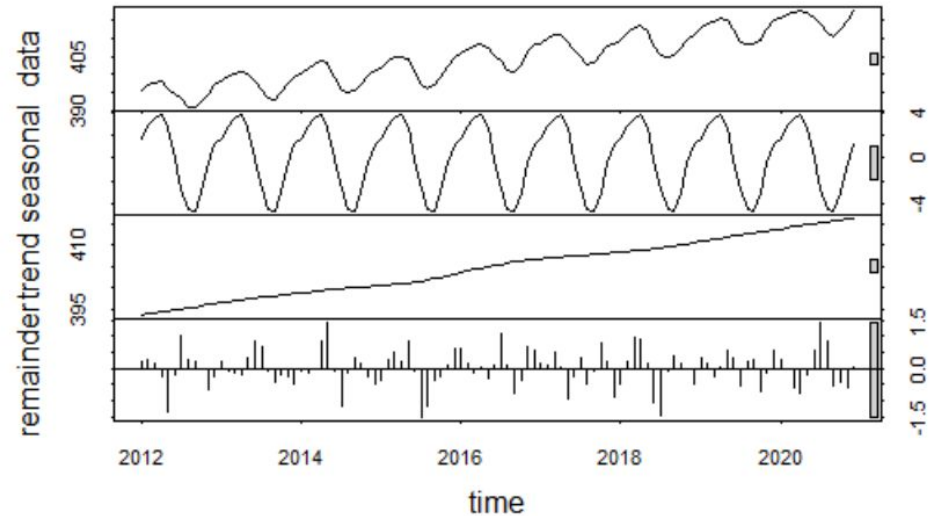
We will have the following steps for the Time Series Regression process:

- ❑ Decomposition
- ❑ Modeling
- ❑ Prediction

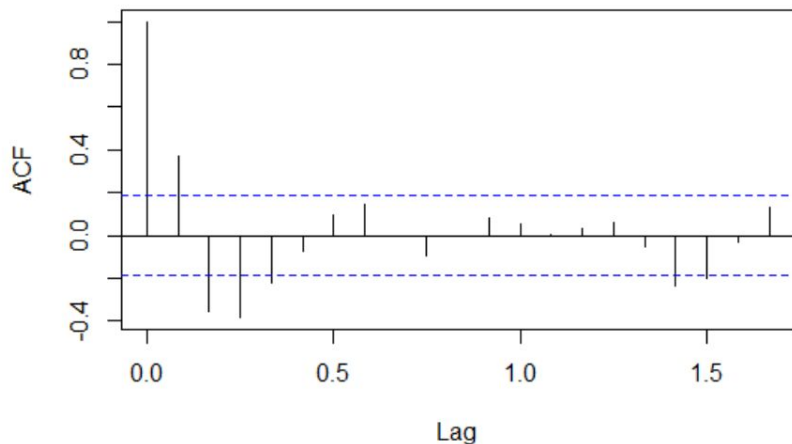
Decomposition

Here is a decomposition of the data and each of the three components modeled separately. It clearly shows that the trend is pretty close to linear and increases over time. The seasonal variation is relatively constant over time. It most likely to use an additive model.

Now let's first model the residuals.



Model the Residuals



- The ACF plot clearly shows the correlation between the residuals because the lags 1,2,3,4 and 14 are outside of the significance bonds.

- There are 4 orders have been selected. AR(4) is going to be used.

Call:

```
ar.yw.default(x = res, order.max = NULL)
```

Coefficients:

1	2	3	4
0.6569	-0.8645	0.3229	-0.3994

Order selected 4 sigma² estimated as 0.1582

Model the Residuals(continue)

- The summary shows the first 4 orders have been removed.
- The AIC value of the model without the mean intercept is 110.8418, which is less than 112.8207, the model with the mean intercept.
- The model without mean intercept has been selected.

Time Series:

Start = 1

End = 6

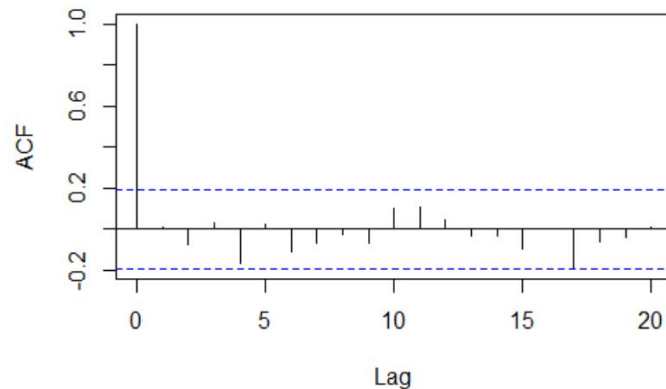
Frequency = 1

[1] NA NA NA NA -0.9967625 0.5157406

AIC Res Model 1: 112.8207

AIC Res Model 2: 110.8418

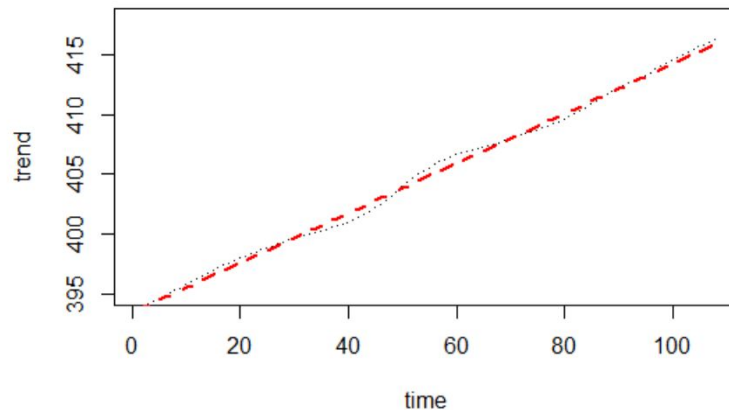
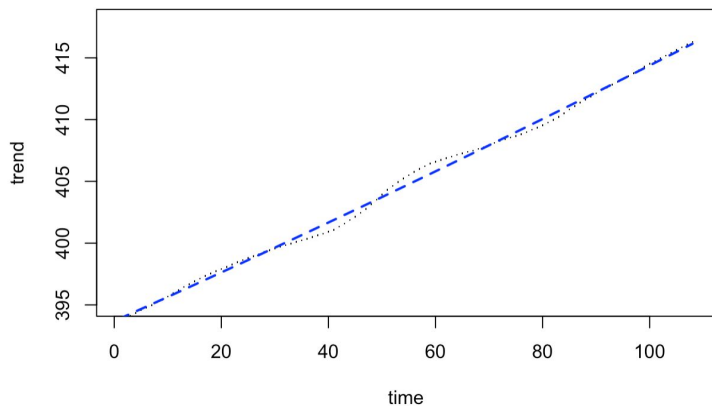
- According to the ACF plot of the AR(4) model, all of the residuals are within the significance bonds.



Model the Trend component

Model1; Trigonometric model

$$\text{trend} = \alpha_1 * \cos(r * \text{time}/12) + \alpha_2 * \sin(r * \text{time}/12) + \text{error}$$



- **Model2: Linear model**

$$\text{trend} = \alpha_0 + \alpha_1 * \text{time} + \text{error}$$

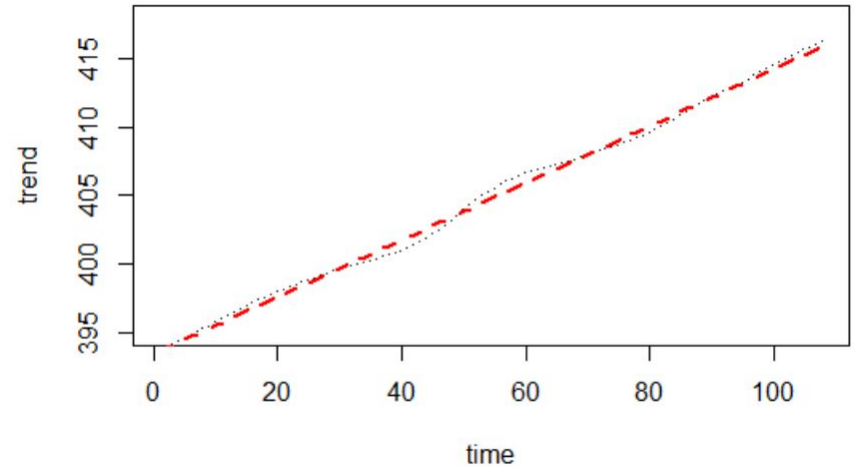
- **There is not significant difference between the two models**

Model the Trend(continue)

The trend tells us the general long-term direction of the data - whether it is increasing or decreasing.

There is a clear upward trend, meaning there's a trend of carbon dioxide levels increasing along with time.

Linear Model: $trend = \alpha_0 + \alpha_1 * time + error$

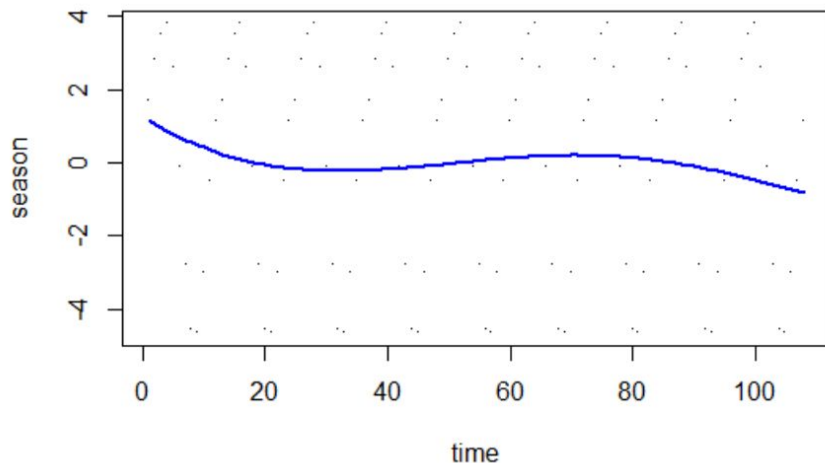


Adjusted R^2 : 0.9963
Residual S.E.: 0.3982

Seasonality

Polynomial Model:

$$\text{fit} = \beta_0 + \beta_1 \text{time} + \beta_2 \text{time}^2 + \beta_3 \text{time}^3 + \varepsilon$$



Adjusted R-squared: -0.004907

Residual standard error: 2.975

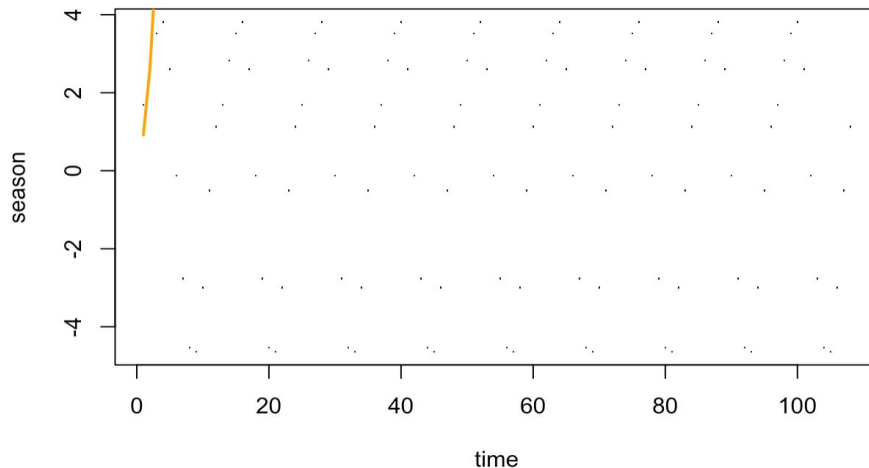
- Seasonality shows the repeated up-and-down cycle of the variable as time goes on.
- Seasonal variation is how the data changes in a specific time frame (monthly in this case).
- Higher R^2 and lower residual error than trigonometric model.

Model the Seasonality(continue)

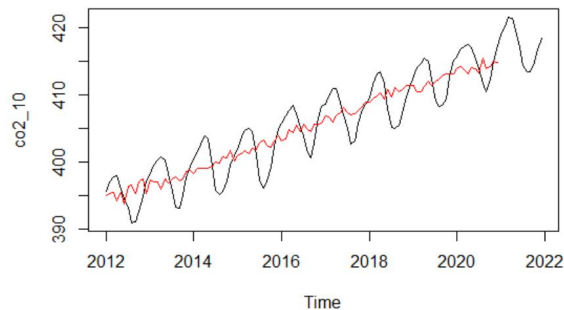
Trigonometric Model: is a pretty odd seasonality model according to the graph

$$fit = \alpha_1 \cos(a * time) + \alpha_2 \sin(a * time) + \alpha_3 \cos(2a * time) + \alpha_4 \sin(2a * time) + \varepsilon$$

where $a=0.017$



Final Models



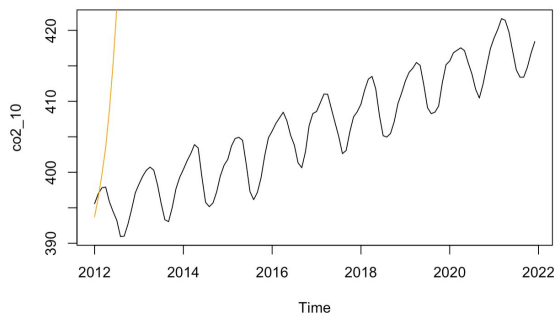
Model 1

$$\text{trend} = \alpha_1 \cos(r * \text{time}/12) + \alpha_2 \sin(a * \text{time}/12) + \varepsilon$$

where $W_t \sim N(0, \sigma^2)$

MSE: 1,248,462

Captured the linear trend of the data but not the seasonality → extremely high mean squared error



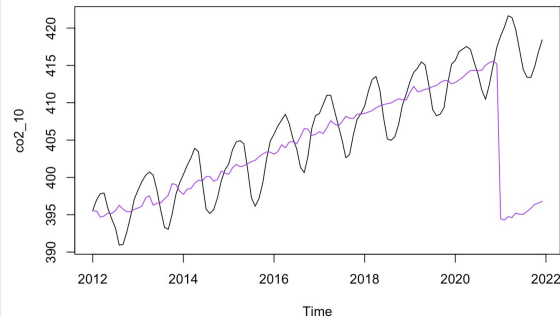
Model 2

$$\text{trend} = a_0 + a_1 * \text{time} + W_t$$

Where $W_t \sim N(0, \sigma^2)$

MSE: 130,175,689

Entirely missed both the trend and seasonality → even higher MSE



Model 3

$$\text{season} = \alpha_1 \cos(a * \text{time}) + \alpha_2 \sin(a * \text{time}) + \alpha_3 \cos(2a * \text{time}) + \alpha_4 \sin(2a * \text{time}) + \varepsilon$$

$W_t = \phi_1 * W_{t-1} + \phi_2 * W_{t-2} + Z_t$, where

$Z_t \sim N(0, \sigma^2)$

MSE: 2,331,641

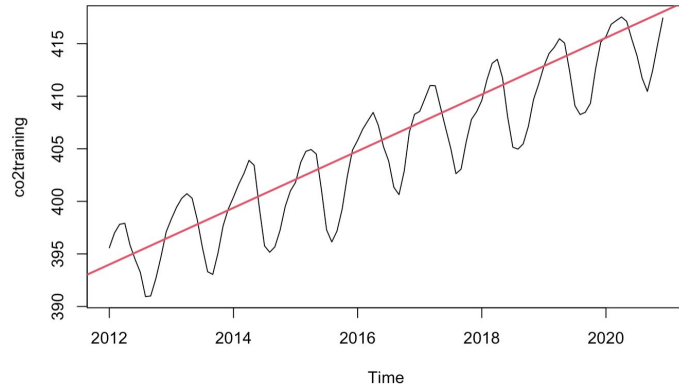
Very similar to Model 1, until 2021 at which the predictions dramatically decreased so it has a higher MSE

*** None of the models are fitting well, we need to try the dummy variable model.**

Model with the Dummy Variables: Model the trend

Linear Trend: *Residual standard error: 3.13*

Adjusted R-squared: 0.8025



Call:

```
lm(formula = co2training ~ t)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-6.5232	-3.0441	0.5343	2.8367	4.5252

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-5032.7538	260.4890	-19.32	<2e-16 ***
t	2.6972	0.1292	20.87	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.13 on 106 degrees of freedom

Multiple R-squared: 0.8043, Adjusted R-squared: 0.8025

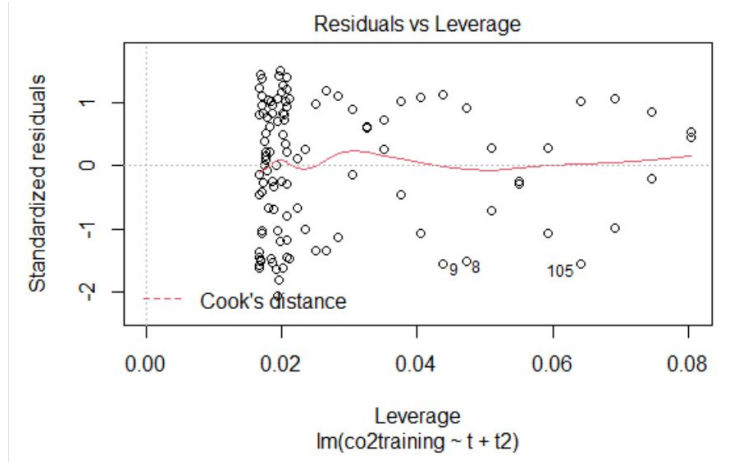
F-statistic: 435.7 on 1 and 106 DF, p-value: < 2.2e-16

Model with the Dummy Variables:

Model the trend (continue)

Quadratic Trend: *Residual standard error: 3.142*

Adjusted R-squared: 0.8009



Call:

```
lm(formula = co2training ~ t + t2)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.404	-3.107	0.648	2.827	4.650

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.594e+04	2.529e+05	0.379	0.705
t	-9.747e+01	2.509e+02	-0.388	0.698
t2	2.484e-02	6.223e-02	0.399	0.691

Residual standard error: 3.142 on 105 degrees of freedom

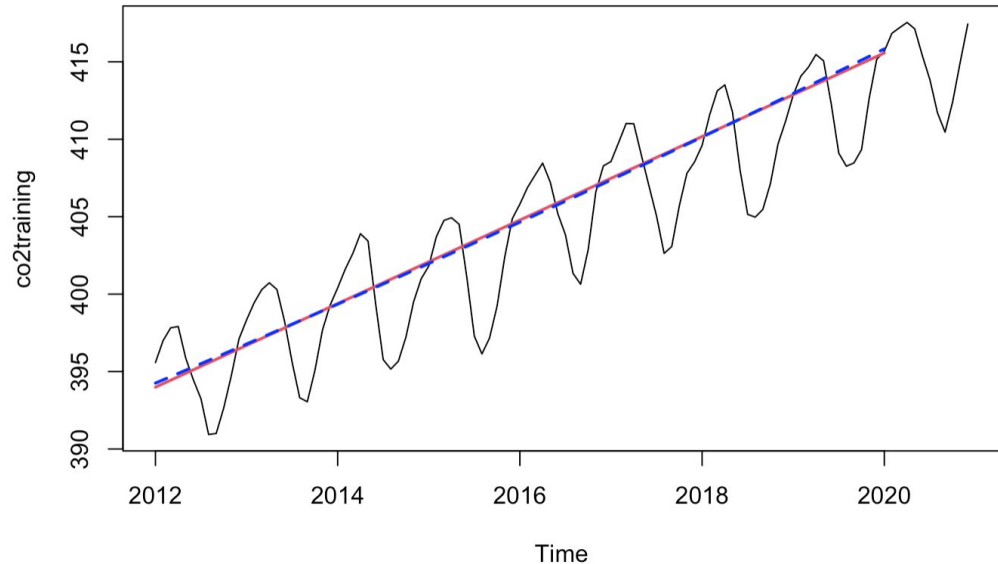
Multiple R-squared: 0.8046, Adjusted R-squared: 0.8009

F-statistic: 216.2 on 2 and 105 DF, p-value: < 2.2e-16

Warning: only using the first two of 3 regression coefficients

Compare the Linear Trend and Quadratic Trend

- ❑ There is a warning in the summary of the the quadratic model, it suggests to use only the first two regression coefficients.
- ❑ According to the graph of both Linear Trend and Quadratic Trend, there is no significant difference between the linear and quadratic trend model.
- ❑ We choose the Linear Trend.

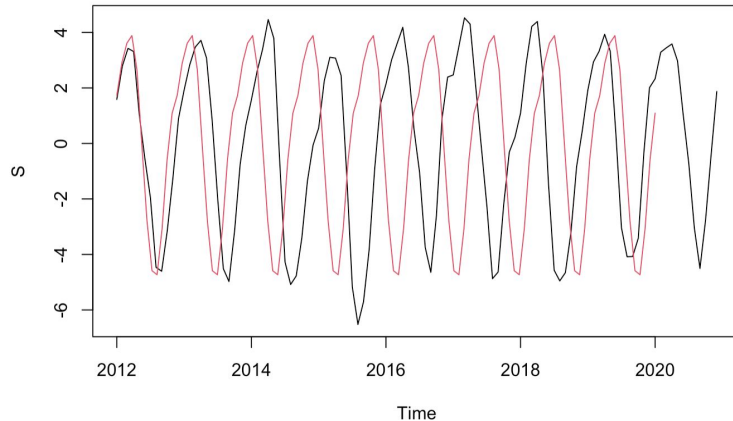


Model with the Dummy Variables:

Model the seasonality with dummy variables

Residual standard error: 0.7601

Adjusted R-squared: 0.9884



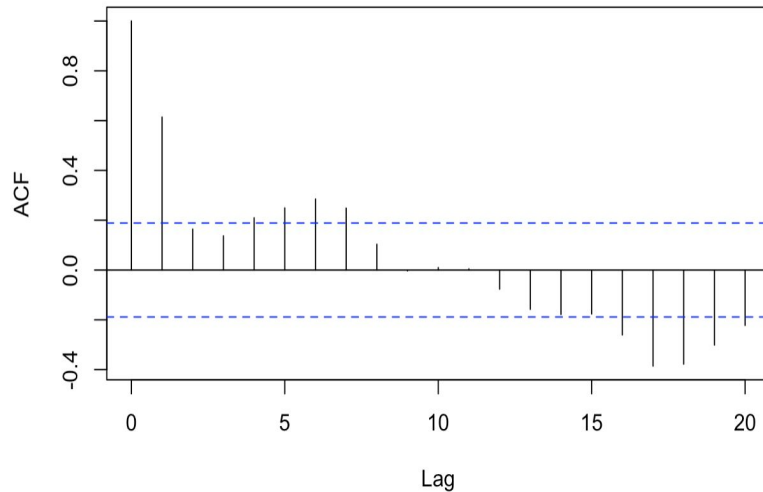
```
Call:
lm(formula = co2training ~ t + M)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-2.30837 -0.36786  0.01397  0.41901  1.75461
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -5.210e+03  6.364e+01 -81.861 < 2e-16 ***
t             2.786e+00  3.157e-02  88.232 < 2e-16 ***
M2            1.156e+00  3.583e-01  3.227  0.00172 **
M3            1.868e+00  3.584e-01  5.212  1.08e-06 ***
M4            2.134e+00  3.584e-01  5.955  4.37e-08 ***
M5            9.024e-01  3.584e-01  2.518  0.01349 *
M6           -1.874e+00  3.585e-01 -5.226  1.02e-06 ***
M7           -4.554e+00  3.586e-01 -12.700 < 2e-16 ***
M8           -6.365e+00  3.587e-01 -17.744 < 2e-16 ***
M9           -6.514e+00  3.588e-01 -18.154 < 2e-16 ***
M10          -4.871e+00  3.589e-01 -13.571 < 2e-16 ***
M11          -2.375e+00  3.591e-01 -6.614  2.19e-09 ***
M12          -7.133e-01  3.593e-01 -1.986  0.04996 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.7601 on 95 degrees of freedom
Multiple R-squared:  0.9897,    Adjusted R-squared:  0.9884
F-statistic: 757.5 on 12 and 95 DF, p-value: < 2.2e-16
```

Model with the Dummy Variables: Model the residuals



As it shows to the left, the indicator model residual was not stationary, so we modeled the residuals by using AR(5).

Call:

```
ar.yw.default(x = res_pre.fit, order.max = NULL)
```

Coefficients:

1	2	3	4	5
1.0242	-0.8585	0.6810	-0.3736	0.2699

Order selected 5 sigma² estimated as 0.2313

AIC Res Model 1: 112.8207

AIC Res Model 2: 110.8418

Model with the Dummy Variables: Model the residuals(continue)

It attained a 2 value of 0.2313. The resulting autocorrelation function can be seen on the following page, vertical lines below the dotted blue lines indicating correlation between certain lag points are not significantly different to zero.

```
Call:  
ar.yw.default(x = res_pre.fit, order.max = NULL)
```

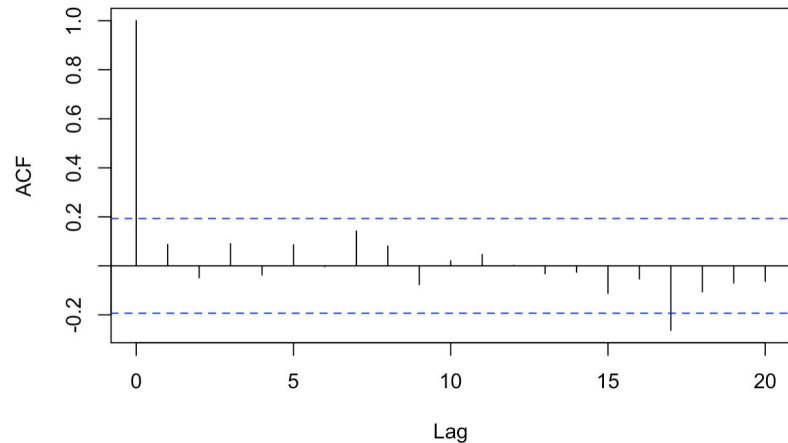
Coefficients:

1	2	3	4	5
1.0242	-0.8585	0.6810	-0.3736	0.2699

```
Order selected 5  sigma^2 estimated as  0.2313  
AIC Res Model 1: 112.8207  
AIC Res Model 2: 110.8418
```

Model with the Dummy Variables: Model the residuals(continue)

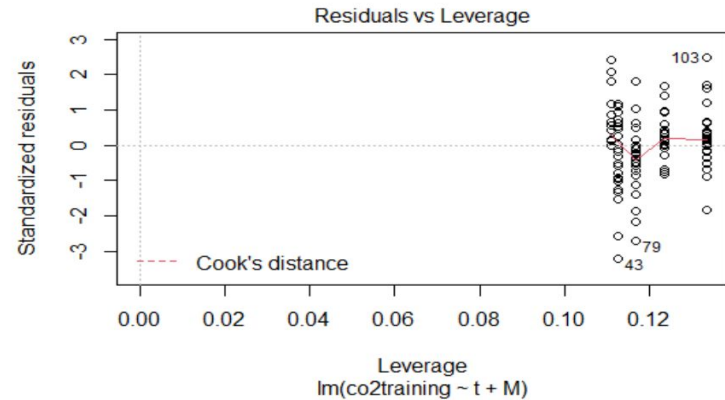
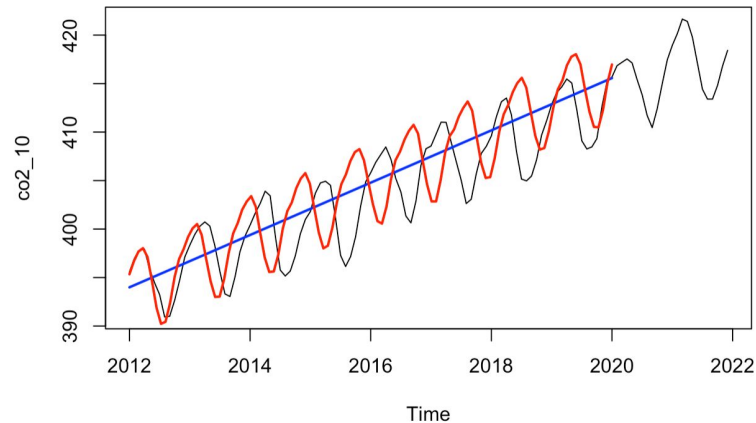
According to the AR summary, 5 orders have been selected, and AIC value for the residual without intercept is 110.8418 (smaller than the one with the intercept). From the ACF, correlogram shows that the sample autocorrelation for the in-sample forecast errors at lag 17 exceeds the significance bounds.



Model with the Dummy Variables: Trend, Seasonality and residuals together

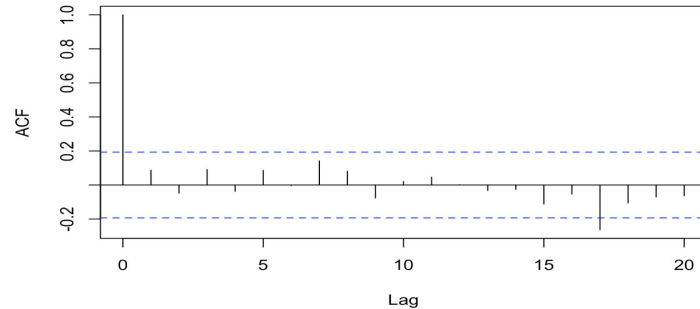
Residual standard error: 0.7491

Adjusted R-squared: 0.8725



Model with the Dummy Variables: Box test

Shown in figure to the right, the p-value of the box test is 0.4393, which is much greater than the critical value 0.05, indicating that there is little evidence of non-zero autocorrelations in the in-sample forecast errors at lags 1-20. Therefore, we can conclude that the indicator model is an adequate predicted model for the CO2 data set.



Box-Ljung test

```
data: resPrefit_ts  
X-squared = 20.3, df = 20, p-value = 0.4393
```

Model with the Dummy Variables: Prediction

The following are results from predicting of 2021 to obtain error statistics and forecasting of 2022 to obtain future values from time series regression. The tables below show the original CO2 values of 2021 from our dataset and the predicted values for 2021, respectively. These were used to calculate the error statistics provided in the table above.

	CO2 <dbl>	t <dbl>	M <dbl>
109	418.94	2021.000	1
110	420.15	2021.083	2
111	421.65	2021.167	3
112	421.41	2021.250	4
113	419.79	2021.333	5
114	417.10	2021.417	6
115	414.42	2021.500	7
116	413.40	2021.583	8
117	413.40	2021.667	9
118	414.81	2021.750	10
119	416.81	2021.833	11
120	418.42	2021.917	12

	fit	lwr	upr
109	417.8992	416.2839	419.5145
110	419.2637	417.6484	420.8790
111	420.1837	418.5684	421.7990
112	420.6581	419.0428	422.2734
113	419.6348	418.0195	421.2501
114	417.0670	415.4517	418.6823
115	414.5948	412.9795	416.2101
116	412.9926	411.3773	414.6079
117	413.0515	411.4362	414.6668
118	414.9026	413.2873	416.5179
119	417.6070	415.9917	419.2223
120	419.4770	417.8617	421.0923

MSE, MAE, MAPE (TSR)

Month	Fitted 2021 CO2(monthly) X_t - hat	2021 original CO2 X_t	$ X_t - \hat{X}_t $	$(X_t - \hat{X}_t)^2$	$(X_t - \hat{X}_t)/X_t$
1	417.8992	418.94	1.0408	1.08326464	0.00248436530290734
2	419.2637	420.15	0.8863	0.78552769	0.00210948470784244
3	420.1837	421.65	1.4663	2.15003569	0.00347752875607732
4	420.6581	421.41	0.7519	0.56535361	0.00178424811940865
5	419.6348	419.79	0.1552	0.02408704	0.000369708663855737
6	417.0670	417.10	0.033	0.001089	0.0000791177175737233
7	414.5948	414.42	0.1748	0.03055504	-0.00042179431494619
8	412.9926	413.40	0.4074	0.16597476	0.000985486211901306
9	413.0515	413.40	0.3485	0.12145225	0.000843009192065796
10	414.9026	414.81	0.0926	0.00857476	-0.000223234733974591
11	417.6070	416.81	0.797	0.635209	-0.00191214222307526
12	419.4770	418.42	1.057	1.117249	-0.00252616987715692
Summation			7.2108	6.68837248	0.00704960752247935
MAE			0.6009		
MSE				0.55736	
MAPE					0.0005875

Applying the original carbon dioxide values of 2021 and the predicted carbon dioxide values of 2021, we can calculate the MAE, MSE, and MAPE for our model, they are approximate 0.6009, 0.5574 and 0.059% respectively.

Exponential Smoothing

Exponential Smoothing was used to forecast the monthly carbon dioxide levels in 2022.

Out of the 3 types of Exponential Smoothing, we decided to implement the Holt-Winters Exponential Smoothing since the time series shows an increasing trend with seasonality.

Estimates 3 parameters: α (alpha), β (beta), and γ (gamma).

Parameters for Exponential Smoothing

Smoothing is controlled by the following parameters:

α (alpha): estimates the level at the current time point

β (beta): estimates the trend component of the time series

γ (gamma): estimates the seasonal component of the time series

All of the parameters above have values that are in between 0 and 1.

Values near 0 indicate that the estimates are based on the recent observation of the time series.

Exponential Smoothing

Holt-Winters exponential smoothing with trend and additive seasonal component.

```
Call:
Holtwinters(x = ten_yrs)
```

Smoothing parameters:

```
alpha: 0.9073384
beta : 0
gamma: 1
```

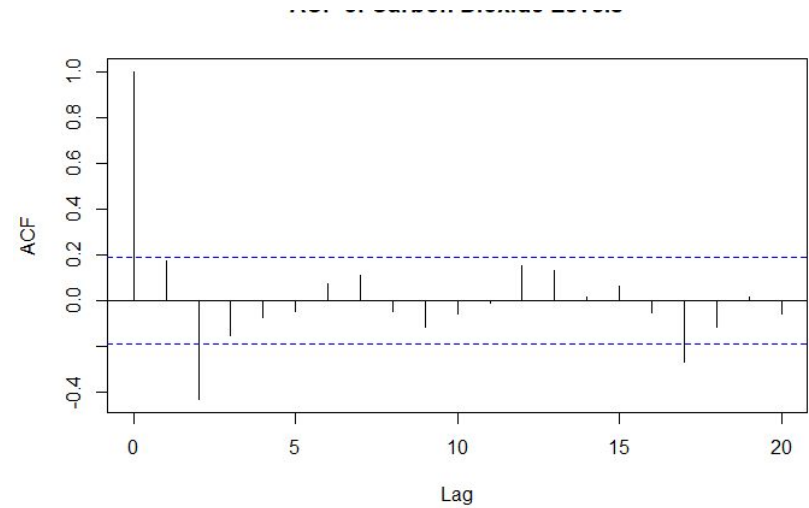
Coefficients:

```
      [,1]
a  417.7711103
b    0.2377972
s1   1.5850055
s2   2.6808683
s3   3.3626216
s4   3.4844321
s5   2.5156537
s6   0.1092531
s7  -2.2763313
s8  -4.0637541
s9  -4.4637133
s10 -2.9880399
s11 -0.8653908
s12  0.6488897
```

- ❑ The high values of α and γ indicate that the estimates of the level and seasonal component depends on the recent observations of the time series.
- ❑ The β value indicates that the slope estimate of the trend component is not updated through the time series.

Exponential Smoothing - Checking Residuals

- ❑ The figure to the right is the ACF plot of the in sample forecast errors. Based on the plot, the sample autocorrelation for lags 2 and 17 are outside of the significance bounds, indicating that there is little non-zero correlation in the residuals.
- ❑ Below the plot is the output from the Box Test. The p-value is 0.0002805, which is smaller than the 5% significance level indicating that there is significant evidence of non-zero correlations in the residuals
- ❑ Based on the residuals, this method may not be adequate in predicting the CO2 levels.



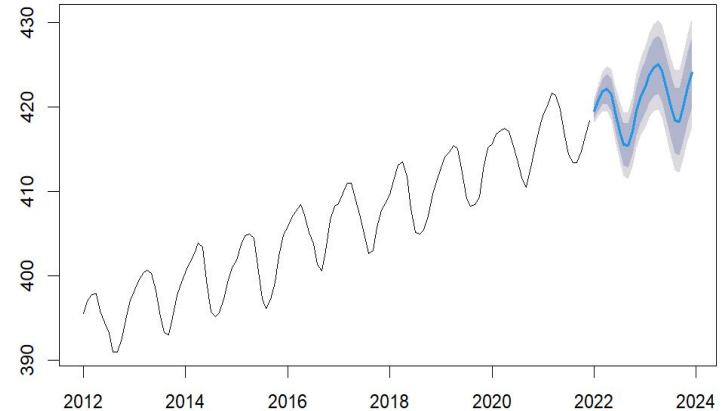
Box-Ljung test

data: newresid
X-squared = 49.281, df = 20, p-value = 0.0002805

Forecasted Values from Exponential Smoothing

	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan 2022	419.5939	418.6500	420.5378	418.1504	421.0374
Feb 2022	420.9276	419.6531	422.2021	418.9784	422.8767
Mar 2022	421.8471	420.3116	423.3826	419.4988	424.1955
Apr 2022	422.2067	420.4486	423.9649	419.5178	424.8956
May 2022	421.4758	419.5201	423.4314	418.4848	424.4667
Jun 2022	419.3071	417.1722	421.4421	416.0420	422.5723
Jul 2022	417.1594	414.8590	419.4597	413.6413	420.6774
Aug 2022	415.6097	413.1552	418.0643	411.8558	419.3637
Sep 2022	415.4476	412.8479	418.0473	411.4717	419.4234
Oct 2022	417.1610	414.4239	419.8982	412.9750	421.3471
Nov 2022	419.5215	416.6535	422.3895	415.1353	423.9077
Dec 2022	421.2736	418.2805	424.2667	416.6960	425.8511

Fig. 2 - Forecasts of Carbon Dioxide Levels in 2022



The table above is the forecasted values of the CO2 levels for 2022. The table shows the 80% and 95% prediction intervals for the CO2 levels. The graph to the right plots the forecasted CO2 levels for 2022 along with past CO2 levels from 2012-2021.

Non-Seasonal ARIMA

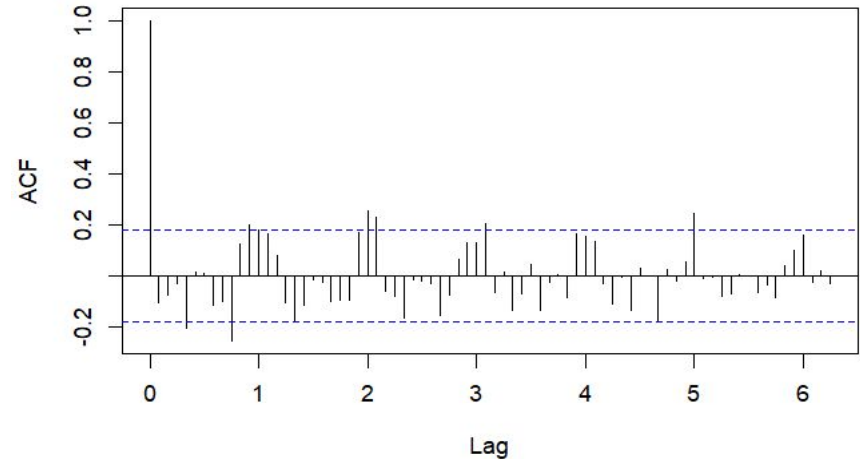
First we used auto ARIMA without the seasonal component to forecast the monthly carbon dioxide levels in 2022.

There are 3 parameters in ARIMA:

1. **AR(p)** stands for autoregressive, the p parameter indicates the number of autoregressive terms.
2. **I(d)** is the differencing part, the d parameter indicates the number of non-seasonal differences needed for stationarity.
3. **MA(q)** stands for moving average, the q is the number of lagged forecast error terms in the prediction equation.

Non-Seasonal ARIMA: Checking Residuals

- ❑ The acf correlogram of the residuals showed that the model had exceeded the significance bounds at multiple lag points.
- ❑ The p-value for the box test was also incredibly low (0.0009471).
- ❑ Which means the forecasted values from model will likely perform poorly

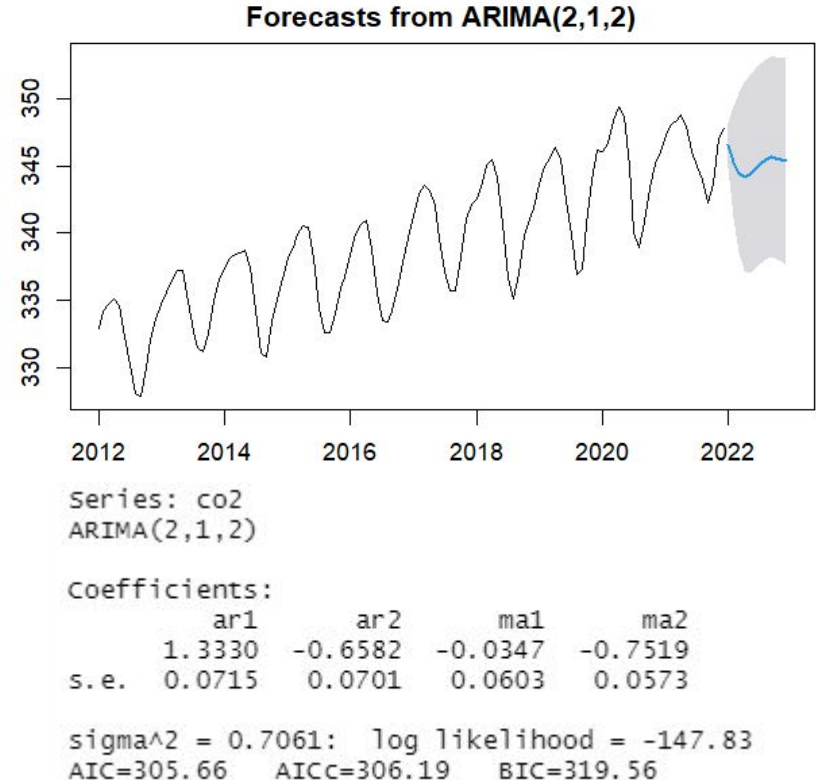


Box-Ljung test

```
data: arimaco2.nonseasonal$residuals  
x-squared = 45.488, df = 20, p-value = 0.0009471
```

Non-Seasonal ARIMA: Forecasting

- ❑ The blue line is the point forecast at each month and the gray area surrounding it is the forecast range using a 95% prediction interval.
- ❑ The forecast for CO2 in 2022 using the non-seasonal ARIMA does not match the previous years pattern very well
- ❑ Using different criteria to select a model such as “bic” or “aic” had no impact on the parameters



Seasonal ARIMA

Finally we used an ARIMA model with the seasonal components included to forecast the monthly carbon dioxide levels in 2022.

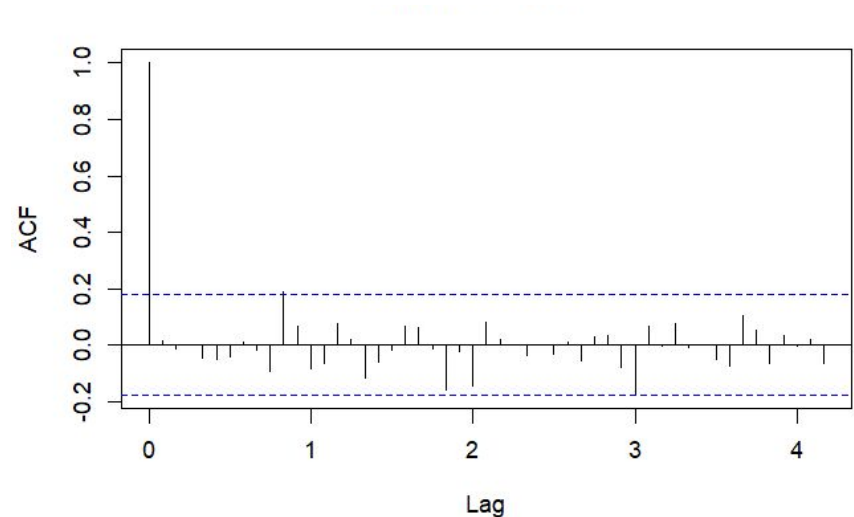
The seasonal ARIMA three additional parameters along with the (p,d,q) parameters from non-seasonal ARIMA:

1. **SAR(P)** stands for the number of seasonal autoregressive terms,
2. **I(D)** is the number of seasonal differences
3. **SMA(Q)** stands for the number of seasonal moving average terms

We use a seasonal ARIMA model because the seasonal pattern of the data is strong and stable over time which prevents the seasonal pattern from "dying out" in long-term forecasts.

Seasonal ARIMA: Checking Residuals

- ❑ The acf correlogram of the residuals showed that the model had stayed within the significance bounds.
- ❑ The box test gave us a very high p-value (0.857).
- ❑ Which meant there was very little evidence for non-zero autocorrelations in the forecast errors.
- ❑ Thus we can conclude that this model will be much better at forecasting the CO2 values for 2022

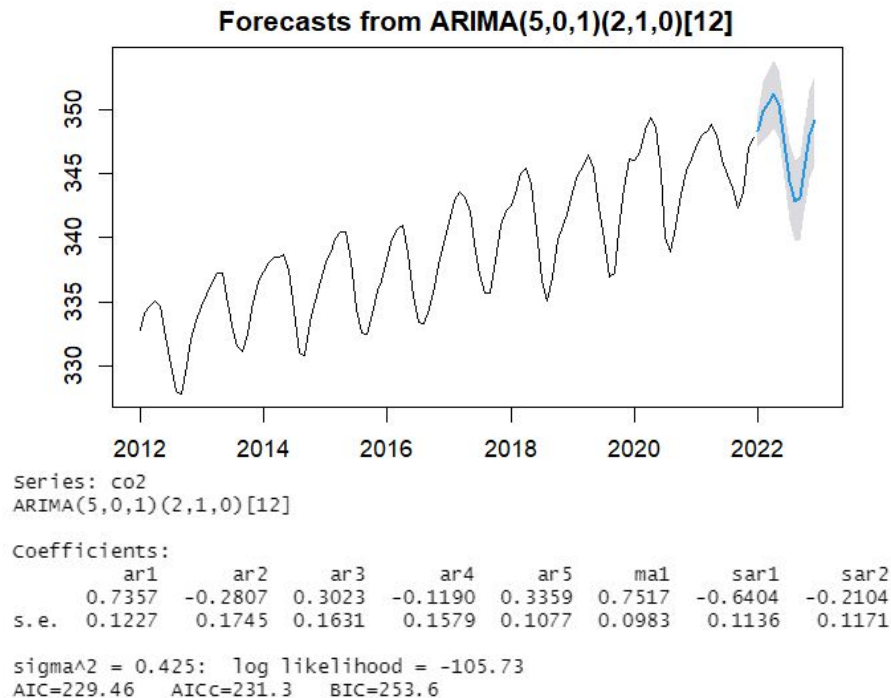


Box-Ljung test

```
data: arimaco2.seasonal$residuals  
x-squared = 13.456, df = 20, p-value = 0.857
```

Seasonal ARIMA: Forecasting

- ❑ The blue line is the point forecast at each month and the gray area surrounding it is the forecast range using a 95% prediction interval.
- ❑ The forecast for CO2 in 2022 using the seasonal ARIMA matches the previous years pattern fairly well
- ❑ Using different criteria to select a model such as “bic” or “aic” had no impact on the parameters



Error Statistics

Calculated by predicting known 2021 carbon dioxide values. Regression statistics calculated using fitted equation through Excel.

Statistic	Regression	Exp. Smoothing	ARIMA
Mean Absolute Error	0.601	1.059	1.262
Mean Squared Error	0.557	1.471	2.979
Mean Percent Error	0.058	-0.19%	-0.004%

Conclusion

- All forecasting methods seem to be reliable
 - Very low error statistics
 - Reasonable projections for future
- Carbon dioxide levels expected to continue rising in the future, keeping its past seasonality patterns
- Time series regression and ARIMA useful in their interpretability
 - Creation of a fitted model with coefficient estimates (contrary to exp. smoothing)
- White noise residuals a major concern for some models
 - Failed Ljung-Box test for exponential smoothing
 - Possible existence of leftover pattern, but results from methods used seem to be reasonable and accurate
- Time series regression may be chosen as most useful model
 - Fitted equation used for forecasting future values
 - Lowest error statistics
 - Potential weakness: reliance on iterative development, infinite potential models exist

References

Any sources, R codes

https://gml.noaa.gov/dv/data/index.php?category=Greenhouse%2BGases&frequency=Monthly%2BAverages¶meter_name=Carbon%2BDioxide

<https://www.c2es.org/content/international-emissions/#:~:text=Globally%2C%20the%20primary%20sources%20of,72%20percent%20of%20all%20emissions>.

https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Carbon_dioxide.